

Trial protocol

Official Title: The Assessment of Educational and Supportive Care to the Depressed Infertile Females Undergo In Vitro Fertilization Procedure by Clinical Pharmacist: Randomized Clinical Trial

Study Description

Brief Summary

Infertility affects married adults, and In Vitro Fertilization (IVF) is an Assisted Reproductive Technology (ART) that can be treated. Women undergoing IVF are more likely to experience depression. There is a need to reduce depression by supporting and advising sufferers. The aim of this study is to evaluate pharmacist counseling's impact on pregnancy rates in depressed infertile females during IVF treatment.

Detailed Description

Infertility affects 10-15% of couples, both men and women, and can cause anxiety, worry, grief, and disappointment. It is a major life experience that can affect all aspects of a couple's life. Preconception counseling is a necessary prerequisite for infertility treatment, as it evaluates the medical, social, genetic, environmental, and occupational factors that influence fertility and pregnancy health. Depression is a common issue among infertile females, and pharmacists play a crucial role in the care of depressed patients. The etiology of female infertility is unknown, but ovulatory disorders, endometriosis, pelvic adhesions, tubal blockage, other tubal abnormalities, and hyperprolactinemia account for 81% of all cases. Age has a negative impact on fertility, and the prevalence of age-related infertility rises as reproductive years decrease. Treatment for infertility aims to resolve treatable causes and overcome intractable ones. General management principles for infertile couples include determining the cause and treatment, suggesting changes in behavior that increase fertility, stopping smoking, and limiting exposure to harmful substances. Assisted reproductive technology (ART) is used to treat infertility, which involves fertility therapies involving both eggs and sperm. In vitro fertilization (IVF) is widely recognized as the most effective method of ART. IVF is one of the most successful therapies for infertility, but it sometimes requires multiple attempts at oocyte picking and embryo transfer. It is effective in treating undiagnosed infertility, obstructed or broken fallopian tubes, genetic disorders, women

with uterine fibroids, ovulation problems, or early ovarian failure, and women who have had their ovaries surgically removed. The IVF cycle involves ovarian stimulation, which involves fertility drugs to stimulate egg production and regulate ovulation. Examples of ovarian stimulation regimens include leuprolide acetate, also known as Lupron, which is used in conjunction with birth control pills for the first week. Anti-estrogens like clomiphene citrate, also known as Clomid, are used to manage anovulation by selectively modulating estrogen receptor activity. Follicle-stimulating hormone (FSH) is a medication that stimulates the ovaries, potentially serving as an alternative therapeutic option for women with anovulation who have undergone unsuccessful cycles of anti-estrogens. Human chorionic gonadotropin injections (HCG) are commonly used to induce ultimate maturation, stimulating ovulation when there is at least one fully developed ovarian follicle. The American Society for Reproductive Medicine advises the transfer of a specific quantity of embryos, whether they are in a fresh or frozen state. Emotional changes in IVF patients are common, with anxiety and depression being more common in infertility patients with psychiatric illness, miscarriage, a longer infertility timeline, and a definitive medical diagnosis. Stress levels increase as therapy becomes more intensive and lasts longer, making IVF patients more stressed than women at the beginning of their infertility examination. Infertility and its medications can cause emotional distress for patients, especially those who have tried multiple IVF cycles without success. Patients may lack familiarity with gonadotropin self-injection, hormone supplements, and other necessary treatments for ovulation induction cycles. Studies show that depression lowers IVF pregnancy rates, but the impact on emotional distress is uncertain. Grief responses are frequently observed among couples experiencing infertility, but typical grief responses can progress into pathological grief, resulting in notable melancholic symptoms. In women, the prevalence of infertility depression ranges from 10-25%, with women experiencing major depression at rates two to three times higher than men. Depression can negatively impact treatment, follow-up, and confidence for the future, as well as the intensity and durability of the affected couple's relationship. The Centre for Epidemiological Studies Depression Scale (CES-D) measures depression symptoms across various demographics, including race, ethnicity, gender, and age. The Fertility Quality of Life Tool (FertiQoL) is an internationally validated instrument designed to measure the quality of life in people experiencing infertility. The FertiQoL questionnaire is beneficial for infertile patients due to its strong construct validity and reliable psychometric properties. Pharmacists play a significant role in education and supportive care for

depressed patients, providing advice, recommendations, and counseling about medications, monitoring patients for drug-related problems, and assessing patient adherence. The modern pharmacist's responsibilities include pharmaceutical care principles, transforming them from a traditional medication seller into a critical member of the healthcare team. The goal of these patient centered interventions is to improve therapeutic outcomes by recognizing, preventing, and treating drug-related problems, encouraging the correct use of medication, and promoting and educating about general health. Pharmaceutical care is a systematic, complete approach that requires pharmacists to work with the healthcare team to detect and treat drug-related difficulties and safeguard patients from medicine-related damage. The idea of care focuses on developing the pharmacist-patient connection and adding value to the therapeutic outcome by actively participating in the treatment method. Evidence-based medicine (EBM) is a practice that combines the results of recent scientific studies with the input of clinicians and patients. The shift in pharmacy practice from perception to evidence-based practice is crucial for optimal pharmacological treatment. The Institute of Medicine aims to achieve 90% of healthcare decisions backed by accurate, latest clinical information by 2020. Collaborative counseling has been shown to reduce stress levels in infertile women receiving IVF by 3.6 times compared to the control group. Psychotherapy is an appealing choice in terms of treatment, as it has been shown to increase patients' chances of becoming pregnant. In most countries, psychological counseling or interpersonal therapy and behavioral activation are the first-line treatments for depression. These studies have shown that infertile women who want to undergo IVF may fail and not get pregnant due to the patient's depression condition prior to the operation. As a result, health education and awareness must be undertaken, and a satisfactory condition must be provided prior to the operation, with a focus on women because they are more susceptible to this depression than men. The key issue is depression in infertile women, which has an impact on the pregnancy rate outcome. There is a need to reduce depression by supporting and advising sufferers. Aim of study This study examines the effect of depression on the pregnancy rate of infertile ladies during IVF procedures and the use of counseling and supporting care by pharmacist to evaluate outcomes of IVF cycles, as measured by clinical pregnancy rates recorded after the intervention. The secondary outcome: assessment of both groups completed the CES-D scale and FertiQoL during the start-time and end-time of the IVF procedure.

Study Type: Interventional

Primary Purpose: Treatment

Study Phase: N/A

Interventional Study Model: Parallel Assignment

The study included 75 patients in the intervention group who received an educational and supportive program designed by a clinical pharmacist. The intervention group received support through five visits: at admission, monitoring medication, monitoring ovulation, monitoring harmful effects, providing emotional support, and following up after egg retrieval during embryo transfer.

Number of Arms: 2

Masking: None (Open Label)

Allocation: Randomized

Enrollment: 150

Arms and Interventions

Arms	Assigned Interventions
Experimental: Group 1 The study included 75 patients in the intervention group who received an educational and supportive program designed by a clinical pharmacist. The intervention group received support through five visits: at admission, monitoring medication, monitoring ovulation, monitoring harmful effects, providing emotional support, and following up after egg retrieval during embryo transfer	Behavioral: Educational and supportive care The trial had a total of 75 participants assigned to the intervention group, whereby they were provided with an educative and supporting program specifically developed by a clinical pharmacist. The intervention group was provided with assistance via a series of five visits, which included an initial visit upon admission, subsequent visits for monitoring medication, ovulation, and negative effects, as well as visits dedicated to giving emotional support. Additionally, a

	follow-up visit was conducted after the egg collecting process and embryo transfer
No Intervention: Group 2 The control group included 75 infertile married ladies attending the hospital for the same purpose and managed in the traditional protocol followed by the hospital system. The study assessed the fertility quality of life and depression status of the patients at the beginning and end of the IVF cycle. All patients received frozen embryo transfers (fertilized eggs), not fresh embryos	

Outcome Measures

Primary Outcome Measure

1. Pregnancy rate

The outcome of IVF cycles, as measured by clinical pregnancy rates, was recorded after the intervention

Secondary Outcome Measure:

2. Fertility Quality of Life.

The assessment of Fertility Quality of Life by: FertiQoL during the start and end times of IVF procedure. The scaled scores range is 0 to 100. Higher scores mean higher quality of life

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Depression status

The assessment of depression will be measured by the Center for Epidemiologic Studies Depression Scale (CES-D). Possible range of scores is zero to 60, with the higher scores indicating the presence of more symptomatology

Eligibility

Minimum Age: 18 Years Maximum Age: 45 Years

Sex: Female Gender Based: Yes

To be an infertile lady Accepts Healthy Volunteers: No

Criteria: Inclusion Criteria:

- To be an infertile lady who spent at least one year after marriage failing to get pregnant and who should then undergo the IVF cycle.
- To be within the reproductive age (18-45 years old); the success rates of IVF tend to decline with increasing age.
- To have the ability to read and understand.
- Have depression

Exclusion Criteria:

- Patient with previous failure of I.V.F. procedure.
- Very obese (BMI > 40).
- Patient with psychiatric problems (psychiatric hospital admission, addiction, neurological or other progressive disease, and psychiatric drug use).
- A patient who has a situation that prevents communication (language and hearing problems).

References

Maleki-Saghooni N, Amirian M, Sadeghi R, Latifnejad Roudsari R. Effectiveness of infertility counseling on pregnancy rate in infertile patients undergoing assisted reproductive technologies: A systematic review and meta-analysis. *Int J Reprod Biomed*. 2017 Jul;15(7):391-402. PubMed 29177241

Al-Homaidan HT. Depression among Women with Primary Infertility attending an Infertility Clinic in Riyadh, Kingdom of Saudi Arabia: Rate, Severity, and Contributing Factors. *Int J Health Sci (Qassim)*. 2011 Jul;5(2):108-15. PubMed 23267288

Nkansah N, Mostovetsky O, Yu C, Chheng T, Beney J, Bond CM, Bero L. Effect of outpatient pharmacists' non-dispensing roles on patient outcomes and prescribing patterns. *Cochrane*

Database Syst Rev. 2010 Jul 7;2010(7):CD000336. doi: 10.1002/14651858.CD000336.pub2.
PubMed 20614422